Cluster Analysis of CART-Synthesised Data: A Factorial Simulation of k-Means and Ward's Linkage

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Abstract

Synthetic data allow information to be shared while protecting confidential records, but their usefulness depends on whether analyses of the synthetic data reproduce those of the original data. We investigate this for cluster analysis through a factorial simulation of 144 scenarios crossing separation, number of groups, dimensionality, correlation and distribution family (720,000 evaluations), applying k-means and Ward's linkage to data synthesised by sequential CART. Agreement between the numbers of clusters estimated from synthetic and original data rises with separation. Overall, k-means recovers the original number more often than Ward's method (0.84 vs. 0.81), on skewed Gamma data (0.81 vs. 0.73) but not on Normal data (0.87 vs. 0.89). A bootstrap null shows that this reflects the greater stability of the k-means

estimate, not smaller damage from CART, and that resampling alone reproduces most centroid and dispersion distortion. A real-data case study agrees. Utility should be judged against a resampling reference.

Keywords: Synthetic data, CART synthesis, `synthpop`, k-means, Ward's linkage, Cluster stability, Bootstrap

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Declarations

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Data availability. All data analysed in this study are simulated; they can be regenerated exactly from the configuration file and seeds distributed with the code. [**TO COMPLETE:** add a DOI (e.g. Zenodo/figshare) if the simulated datasets are also deposited]

Code availability. The full pipeline, from data generation through to the figures and tables reported in the article, is available at https://github.com/Sterrysx/synthetic-clustering. [**TO COMPLETE:** archived release with DOI, and the anonymised mirror used for review, e.g. https://anonymous.4open.science/r/XXXX]

Ethics approval and consent. Not applicable: the study uses only simulated data.